

Multimodal Alignment and Fusion: A Survey

Songtao Li · Hao Tang

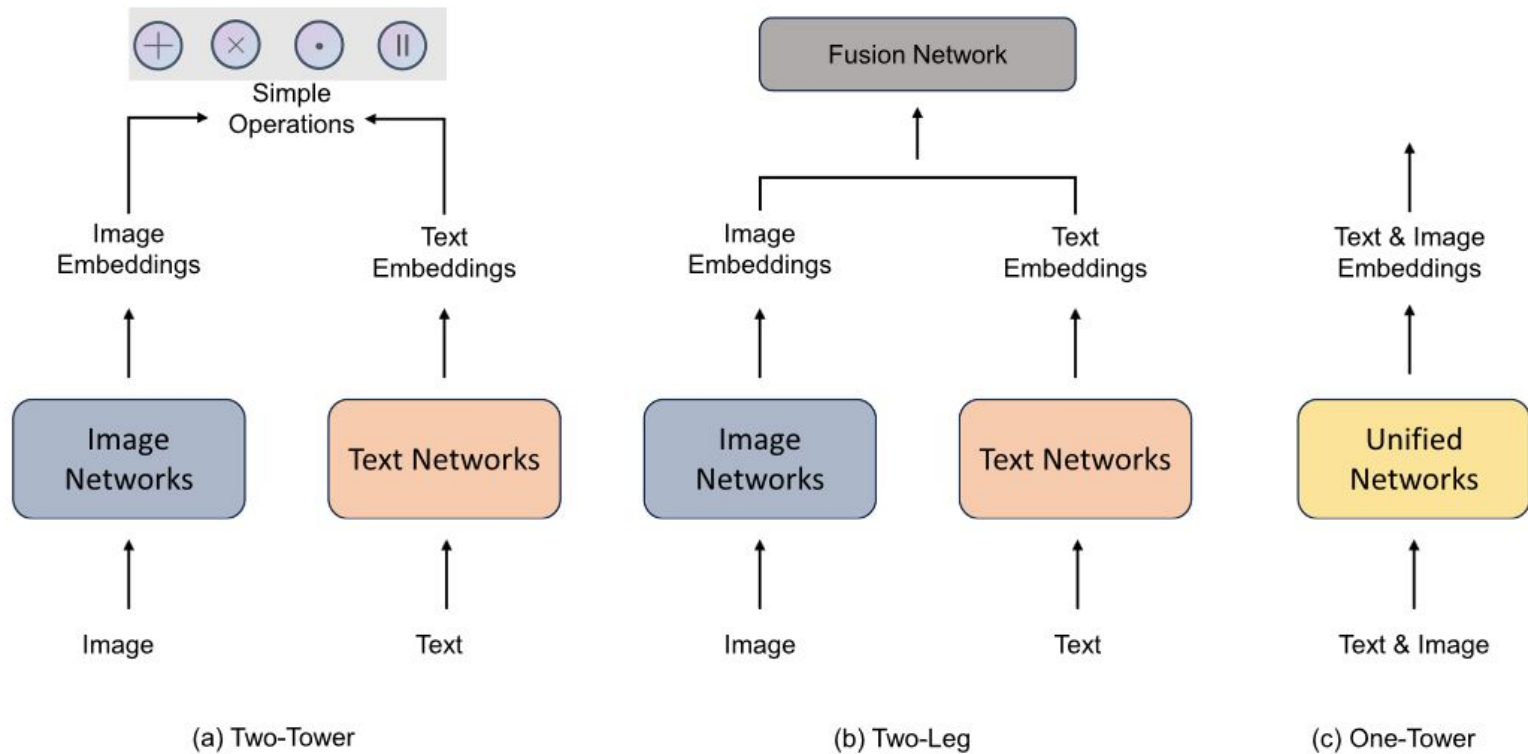


Fig. 1 Overview of multimodal model architectures: (a) Two-Tower (Radford et al., 2021; Jia et al., 2021; Cui et al., 2024; Vasilakis and Bittner, 2024; Xu et al., 2023; Su et al., 2023; Liu et al., 2023a; Chen et al., 2024; Fei et al., 2022; Wen et al., 2024; Tu et al., 2022; Yuan et al., 2021): processes images and text separately, combining embeddings through simple operations (add, multiple, dot product and concatenate); (b) Two-Leg (Allaire and Willcox, 2012; Badrinarayanan and Kendall, 2015; Danapal et al., 2020; Guo and Zhang, 2023; Jaiswal et al., 2015;

Li et al., 2020b; Li and Wu, 2018; Mai et al., 2019; Makris et al., 2011; Missaoui et al., 2010; Rövid and Remeli, 2019; Steinbaeck et al., 2018; Uezato et al., 2020; Wei et al., 2021; Kim et al., 2021): combines separate image and text embeddings using a Fusion Network; (c) One-Tower (Bao et al., 2022; Li et al., 2022, 2023b; Dai et al., 2023; Chen and Xu, 2023; Zhu et al., 2023; Wang and Jiahui, 2022; Wang et al., 2024a; Bai et al., 2023; Alayrac et al., 2022): utilizes a unified network to jointly embed image and text inputs

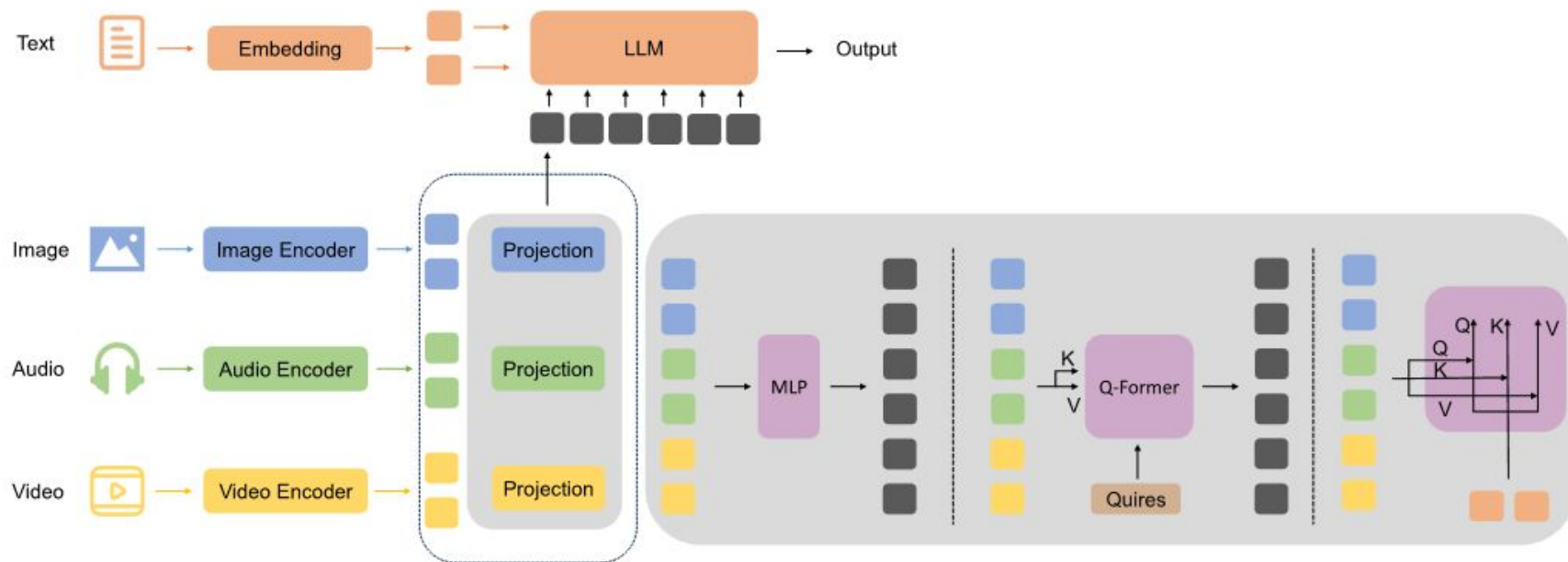
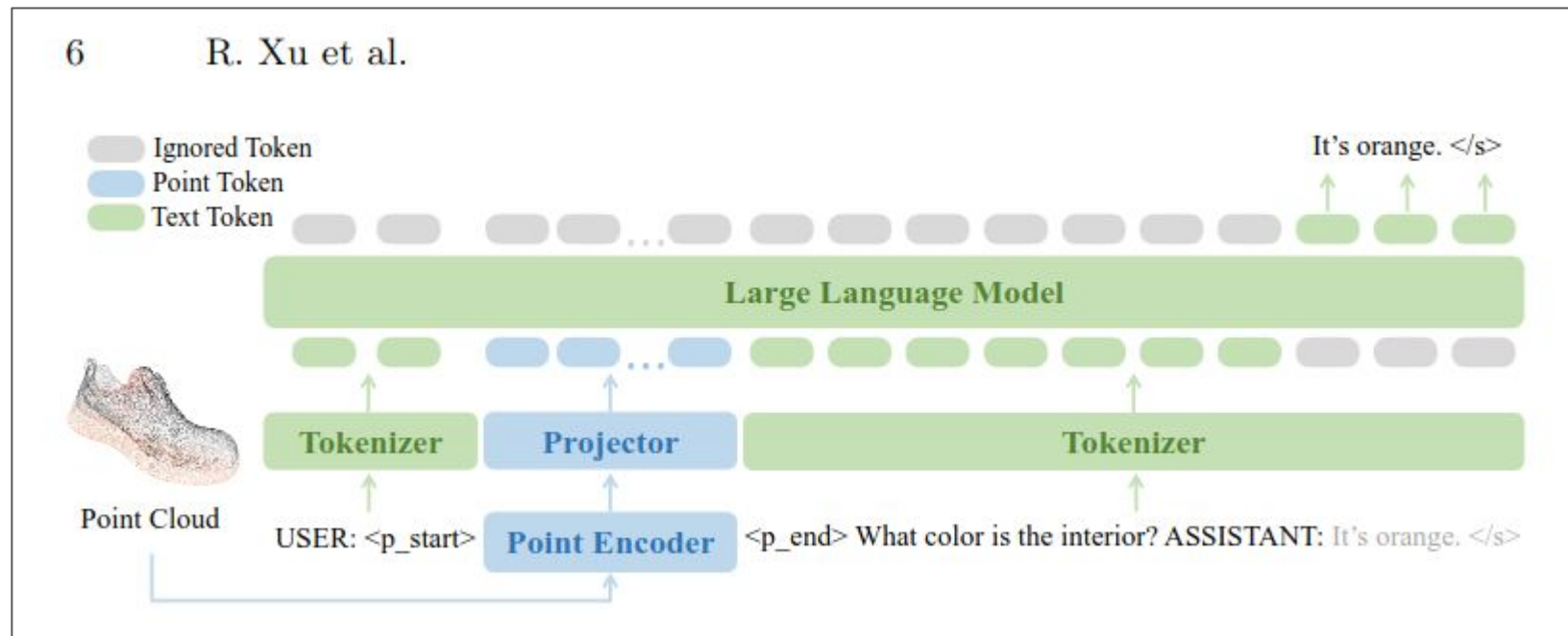


Fig. 7 This pipeline demonstrates multimodal fusion using a large language model (LLM). Text inputs are embedded and processed by the LLM, while image, audio, and video inputs are encoded, projected into a shared embedding space, and passed through modules such as an MLP

and Q-Former. The Q-Former uses attention mechanisms (queries, keys, and values) to align multimodal features before generating a final output through the LLM

PointLlm



Key challenge: aligning point clouds tokens to text token space.